

Report for ForestQuery into Global Deforestation, 1990 to 2016

ForestQuery is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. GLOBAL SITUATION

According to the World Bank, the total forest area of the world was **41282694.9 Sq. KM** in 1990. As of 2016, the most recent year for which data was available, that number had fallen to **39958245.9 sq. KM**, a loss of **1324449 sq. KM**, or ~ **3.2%**.

The forest area lost over this time period is slightly more than the entire land area of **Peru** listed for the year 2016 (which is **1279999.99 sq. KM**).

2. REGIONAL OUTLOOK

In 2016, the percent of the total land area of the world designated as forest was **31.38%**. The region with the highest relative forestation was **Latin America & Caribbean**, with **46.16%**, and the region with the lowest relative forestation was **Middle East & North Africa**, with **2.07%** forestation.

In 1990, the percent of the total land area of the world designated as forest was **32.42%**. The region with the highest relative forestation was **Latin America & Caribbean**, with **51.03%**, and the region with the lowest relative forestation was **Middle East & North Africa**, with **1.78%** forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51.03	46.16
Sub-Saharan Africa	32.19	27.56
World	32.42	31.38

The only regions of the world that decreased in percent forest area from 1990 to 2016 were **Latin America & Caribbean** (dropped from **51.03%** to **46.16%**) and **Sub-Saharan Africa** (**32.19%** to **27.56%**). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from **32.42%** to **31.38%**.

3. COUNTRY-LEVEL DETAIL

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, **China**. This country actually increased in forest area from 1990 to 2016 by **33.55%**. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the **United States**, but it only saw an increase of **79200 sq. KM**, much lower than the figure for **China**.

China and **United States** are of course very large countries in total land area, so when we look at the largest *percent* change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. **Iceland** increased in forest area by **213.66%** from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990 to 2016. The following 5 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change (sq. KM)
Brazil	Latin America & Caribbean	541510
Indonesi	East Asia & Pacific	282193.98
Myanmar	East Asia & Pacific	107234
Nigeria	Sub-Saharan Africa	106506
Tanzania	Sub-Saharan Africa	102320

The second way to consider which countries are of concern is to analyze the data by percent decrease.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-saharan Africa	75.45
Nigeria	Sub-saharan Africa	61.80
Uganda	Sub-saharan Africa	59.27
Mauritania	Sub-saharan Africa	46.75
Honduras	Latin America & Caribbean	45.03

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of **Sub-saharan Africa**. The countries are **Togo**, **Nigeria**, **Uganda**, and **Mauritania**. The 5th country on the list is **Honduras**, which is in the **Latin America & Caribbean** region.

From the above analysis, we see that **Nigeria** is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
00 - 25 %	85
25 - 50 %	73
50 - 75 %	38
75 - 100 %	09

The largest number of countries in 2016 were found in the 00 - 25 % quartile.

There were nine countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
Suriname	Latin America & Caribbean	98.26%
Micronesia, Fed. Sts.	East Asia & Pacific	91.86%
Gabon	Sub-saharan Africa	90.04%
Seychelles	Sub-saharan Africa	88.41%
Palau	East Asia & Pacific	87.61%
American Samoa	East Asia & Pacific	87.50%
Guyana	Latin America & Caribbean	83.90%
Lao PDR	East Asia & Pacific	82.11%
Solomon Islands	East Asia & Pacific	77.86%

4. RECOMMENDATIONS

Write out a set of recommendations as an analyst on the ForestQuery team.

- *What have you learned from the World Bank data?*
 - ***We may need more up-to-date data to analyze and know the latest changes _We are in 2026 but this data maximum year was 2016!_***
 - ***Deforestation is a global issue, we need to investigate and collect more data to know the reasons for the governments and individuals are going in this harmful manner on the environment.***

- *Which countries should we focus on over others?*
 - ***During our research we had found lot of countries had extremely high percent decrease in forest area from 1990 to 2016, We need to focus on them and there were: Togo, Nigeria, Uganda, Mauritania, Honduras, and Pakistan.***
 - ***Also this countries with the largest amount decrease in forest area: Brazil, Indonesia, Myanmar, Nigeria, and Tanzania.***

5. APPENDIX: SQL Queries Used

```
-- Create view "forestation_primary"
CREATE VIEW forestation_primary AS
SELECT
f.country_code AS country_code ,
f.country_name AS country_name ,
f.forest_area_sqkm,
l.total_area_sq_mi,
r.income_group AS income_group,
r.region AS region, f.year
FROM forest_area f
INNER JOIN land_area l
ON f.country_code = l.country_code AND f.year = l.year
INNER JOIN regions r
ON l.country_code = r.country_code;

-- convert mile to km for total_area, create another view "forestation"
CREATE VIEW forestation AS
SELECT
country_code,
country_name,
forest_area_sqkm,
ROUND(total_area_sq_mi*2.59,2) AS total_area_sqkm,
ROUND((forest_area_sqkm/(total_area_sq_mi*2.59))*100,2) AS forest_percentage,
income_group,
region,
year
FROM forestation_primary;
```

Part 1 - Global Situation

a. What was the total forest area (in sq km) of the world in 1990? Please keep in mind that you can use the country record denoted as "World" in the region table.

```
SELECT
(forest_area_sqkm) AS total_forest_area_1990
FROM forestation
WHERE country_name = 'World' AND year = 1990;
```

b. What was the total forest area (in sq km) of the world in 2016? Please keep in mind that you can use the country record in the table is denoted as "World."

```
SELECT
    (forest_area_sqkm) AS total_forest_area_2016
FROM forestation
WHERE country_name = 'World' AND year = 2016;
```

c. What was the change (in sq km) in the forest area of the world from 1990 to 2016?

```
SELECT
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS total_forest_area_1990,
    SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS total_forest_area_2016,
    SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) -
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS
change_in_forest_area_sqkm
FROM forestation
WHERE country_name = 'World' ;
-- AND year IN (1990, 2016);
```

d. What was the percent change in forest area of the world between 1990 and 2016?

```
SELECT
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS total_forest_area_1990,
    SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS total_forest_area_2016,
    ((SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) -
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END))/SUM(CASE WHEN year = 1990
THEN forest_area_sqkm ELSE 0 END))*100.0 AS percentage_change_in_forest_area
FROM forestation
WHERE country_name = 'World'
    AND year IN (1990, 2016);
```

e. If you compare the amount of forest area lost between 1990 and 2016, to which country's total area in 2016 is it closest to?

```
WITH forest_loss AS (
    SELECT
        ABS(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) -
            SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) AS area_lost_sqkm
    FROM forestation
    WHERE country_name = 'World'
        AND year IN (1990, 2016)
),
country_comparison AS (
    SELECT
        f.country_name,
        f.total_area_sqkm AS country_total_area_2016,
        fl.area_lost_sqkm AS forest_area_lost,
        ABS(f.total_area_sqkm - fl.area_lost_sqkm) AS difference,
        RANK( ) OVER (ORDER BY ABS(f.total_area_sqkm - fl.area_lost_sqkm)) AS rank_closest
    FROM forestation f
    CROSS JOIN forest_loss fl
    WHERE f.year = 2016
        AND f.country_name != 'World'
        AND f.total_area_sqkm IS NOT NULL
)
SELECT
    country_name,
    country_total_area_2016,
    forest_area_lost,
    difference
FROM country_comparison
WHERE rank_closest = 1;
```

Part 2 - Regional Outlook

a. What was the percent forest of the entire world in 2016? Which region had the HIGHEST percent forest in 2016, and which had the LOWEST, to 2 decimal places?

```
-- 1
WITH region_table AS (
    SELECT region,
        (SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
            total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
        (SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
            total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
    FROM forestation
    WHERE year IN (1990, 2016)
        AND region IS NOT NULL
```



```

    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT ROUND(percent_forest_area_2016,2) AS percent_forest_area_2016_world
FROM region_table
WHERE region = 'World';

```

```

-- 2
-- Which region had the HIGHEST percent forest in 2016
WITH region_table AS (
SELECT
region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT
    region AS highest_region,
    ROUND(percent_forest_area_2016,2) AS percent_forest_area_2016
FROM region_table
ORDER BY 2 DESC
LIMIT 1;

```

```

-- 3
-- the LOWEST percent forest in 2016
WITH region_table AS (
SELECT region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT

```

```

    region AS lowest_region,
    ROUND(percent_forest_area_2016,2) AS percent_forest_area_2016
FROM region_table
ORDER BY 2 ASC
LIMIT 1;

```

b. What was the percent forest of the entire world in 1990? Which region had the HIGHEST percent forest in 1990, and which had the LOWEST, to 2 decimal places?

```

-- 1
WITH region_table AS (
SELECT region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT
    ROUND(percent_forest_area_1990,2) AS percent_forest_area_1990_world
FROM region_table
WHERE region = 'World';

```

```

-- 2
-- Which region had the HIGHEST percent forest in 1990
WITH region_table AS (
SELECT region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT
    region AS highest_region,

```

```

        ROUND(percent_forest_area_1990,2) AS percent_forest_area_1990
FROM region_table
ORDER BY 2 DESC
LIMIT 1;

```

```

-- 3
-- the LOWEST percent forest in 1990
WITH region_table AS (
SELECT region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016 THEN
total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region
ORDER BY region
)
SELECT
    region AS lowest_region,
    ROUND(percent_forest_area_1990,2) AS percent_forest_area_1990
FROM region_table
ORDER BY 2 ASC
LIMIT 1;

```

c. Based on the table you created, which regions of the world DECREASED in forest area from 1990 to 2016?

```

-- The LOWEST percent forest in 1990

WITH region_table AS (
SELECT region,
(SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)*100.0)/SUM(CASE WHEN year =1990
THEN total_area_sqkm ELSE 0 END) AS percent_forest_area_1990,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)*100.0)/SUM(CASE WHEN year =2016
THEN total_area_sqkm ELSE 0 END) AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND region IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY region

```

```

ORDER BY region
)
SELECT
    region,
    ROUND(percent_forest_area_1990,2) AS percent_forest_area_1990 ,
    ROUND(percent_forest_area_2016,2) AS percent_forest_area_2016,
    ROUND(percent_forest_area_2016 - percent_forest_area_1990,2) AS change_in_percent
FROM region_table
WHERE percent_forest_area_2016 < percent_forest_area_1990
ORDER BY change_in_percent ASC;

```

Part 3 - Country-Level Detail

```

-- SUCCESS STORIES
-- China query
WITH country_fa AS
(
SELECT
    country_name,
    region,
    (SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_1990,
    (SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND country_name IS NOT NULL
    AND country_name != 'World'
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY country_name, region
HAVING SUM(CASE WHEN year = 1990 THEN total_area_sqkm ELSE 0 END) > 0
    AND SUM(CASE WHEN year = 2016 THEN total_area_sqkm ELSE 0 END) > 0
ORDER BY country_name
)
SELECT
    country_name,
    region,
    forest_area_1990,
    forest_area_2016,
    ROUND( - forest_area_1990 + forest_area_2016,2) AS increased_area_sqkm,
    ROUND( ((- forest_area_1990 + forest_area_2016)/forest_area_1990) *100, 2) AS
increased_pct
FROM country_fa
ORDER BY increased_area_sqkm DESC
LIMIT 5;

```

```

-- best increased_pct
-- Iceland query
WITH country_fa AS
(
SELECT
    country_name,
    region,
    (SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_1990,
    (SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND country_name IS NOT NULL
    AND country_name != 'World'
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY country_name, region
HAVING SUM(CASE WHEN year = 1990 THEN total_area_sqkm ELSE 0 END) > 0
    AND SUM(CASE WHEN year = 2016 THEN total_area_sqkm ELSE 0 END) > 0
ORDER BY country_name
)
SELECT
    country_name,
    region,
    forest_area_1990,
    forest_area_2016,
    ROUND( - forest_area_1990 + forest_area_2016, 2) AS increased_area_sqkm,
    ROUND( ((- forest_area_1990 + forest_area_2016)/forest_area_1990) * 100, 2) AS
increased_pct
FROM country_fa
ORDER BY increased_pct DESC
LIMIT 5;

```

a. Which 5 countries saw the largest amount decrease in forest area from 1990 to 2016? What was the difference in forest area for each?

```

WITH country_fa AS
(
SELECT
    country_name,
    region,
    (SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_1990,
    (SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND country_name IS NOT NULL
    AND country_name != 'World'
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY country_name, region

```

```

HAVING SUM(CASE WHEN year = 1990 THEN total_area_sqkm ELSE 0 END) > 0
      AND SUM(CASE WHEN year = 2016 THEN total_area_sqkm ELSE 0 END) > 0
ORDER BY country_name
)
SELECT
    country_name,
    region,
    ROUND(forest_area_1990 - forest_area_2016,2) AS decreased_area_sqkm,
    forest_area_1990,
    forest_area_2016
FROM country_fa
ORDER BY decreased_area_sqkm DESC
LIMIT 5;

```

b. Which 5 countries saw the largest percent decrease in forest area from 1990 to 2016? What was the percent change to 2 decimal places for each?

```

WITH country_fa_per AS
(
SELECT
    country_name,
    region,
    (SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =1990
THEN total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_1990,
    (SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END)/SUM(CASE WHEN year =2016
THEN total_area_sqkm ELSE 0 END))*100 AS percent_forest_area_2016
FROM forestation
WHERE year IN (1990, 2016)
    AND country_name IS NOT NULL
    AND country_name != 'World'
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
GROUP BY country_name, region
HAVING SUM(CASE WHEN year = 1990 THEN total_area_sqkm ELSE 0 END) > 0
      AND SUM(CASE WHEN year = 2016 THEN total_area_sqkm ELSE 0 END) > 0
ORDER BY country_name
)
SELECT
    country_name,
    region,
    ROUND (((percent_forest_area_1990 -
percent_forest_area_2016)/percent_forest_area_1990)*100,2) AS
decreased_percentatge_forest_area,
    percent_forest_area_1990,
    percent_forest_area_2016
FROM country_fa_per
WHERE percent_forest_area_1990 > percent_forest_area_2016
ORDER BY decreased_percentatge_forest_area DESC
LIMIT 5;

```

c. If countries were grouped by percent forestation in quartiles, which group had the most countries in it in 2016?

```
WITH country_forest_pct AS (  
    SELECT  
        country_name,  
        (forest_area_sqkm / total_area_sqkm) * 100 AS percent_forestation  
    FROM forestation  
    WHERE year = 2016  
        AND country_name IS NOT NULL  
        AND forest_area_sqkm IS NOT NULL  
        AND total_area_sqkm IS NOT NULL  
        AND total_area_sqkm > 0  
)  
SELECT  
    CASE  
        WHEN percent_forestation >= 0 AND percent_forestation < 25 THEN '0-25%'  
        WHEN percent_forestation >= 25 AND percent_forestation < 50 THEN '25-50%'  
        WHEN percent_forestation >= 50 AND percent_forestation < 75 THEN '50-75%'  
        WHEN percent_forestation >= 75 THEN '75-100%'  
    END AS quartile_range,  
    COUNT(*) AS country_count  
FROM country_forest_pct  
GROUP BY quartile_range  
ORDER BY country_count DESC;
```

d. List all of the countries that were in the 4th quartile (percent forest > 75%) in 2016.

```
WITH country_forest_pct AS (  
    SELECT  
        country_name,  
        region,  
        ROUND((forest_area_sqkm / total_area_sqkm) * 100, 2) AS percent_forestation  
    FROM forestation  
    WHERE year = 2016  
        AND country_name IS NOT NULL  
        AND forest_area_sqkm IS NOT NULL  
        AND total_area_sqkm IS NOT NULL  
        AND total_area_sqkm > 0  
)  
SELECT  
    country_name,  
    region,  
    percent_forestation  
FROM country_forest_pct  
WHERE percent_forestation > 75  
ORDER BY percent_forestation DESC;
```

e. How many countries had a percent forestation higher than the United States in 2016?

```
-- (94 countries)

WITH country_forest_pct AS (
  SELECT
    country_name,
    region,
    ROUND((forest_area_sqkm / total_area_sqkm) * 100, 2) AS percent_forestation
  FROM forestation
  WHERE year = 2016
    AND country_name IS NOT NULL
    AND forest_area_sqkm IS NOT NULL
    AND total_area_sqkm IS NOT NULL
    AND total_area_sqkm > 0
)
SELECT COUNT(*) AS count_top_countries
FROM country_forest_pct
WHERE percent_forestation > ( SELECT percent_forestation FROM country_forest_pct WHERE
country_name = 'United States' );
```